Compartmental models in mathematical epidemiology

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Definitions

Definition

Epidemiology is the study of the occurrence and distribution of health-related events, states, and processes in specified populations, including the study of the determinants influencing such processes, and the application of this knowledge to control relevant health problems.

Definition

Infectious disease is a process caused by an agent, often a type of microorganism, that impairs a person's health.

Definition

Epidemic is the occurrence in a community or region of cases of an illness, specific health-related behavior, or other health-related events clearly in excess of normal expectancy.

Definition

A *pandemic* is the outbreak of infectious disease that occurs over a wide geographical area and that is of high prevalence, generally affecting a significant proportion of the world's population, usually over the course of several months.

Definition

A *host* is a person or other living animal, including birds and arthropods, that provides subsistence or lodgment to an infectious agent under natural conditions.

Definition

A *susceptible* is a member of a population who is at risk of becoming infected by a disease.

Disease progression

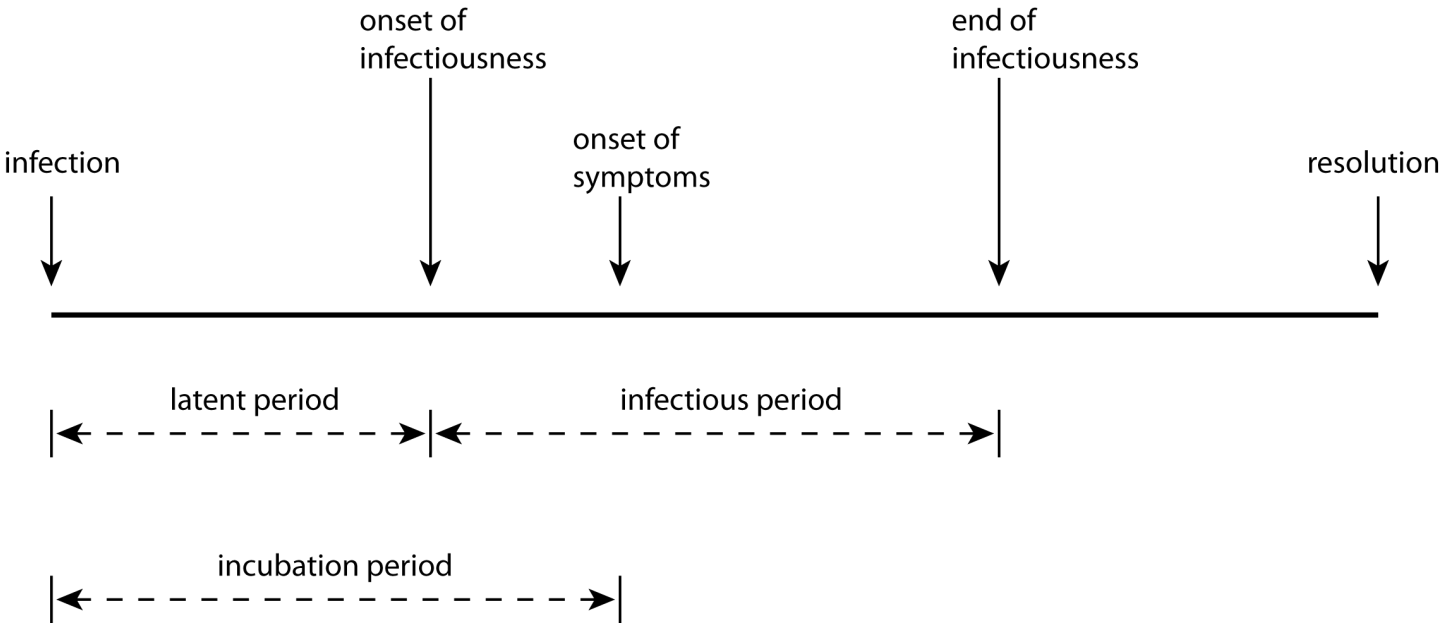

Figure: Transfer diagram for a simple SIR model

The lengths of the latent, incubation, and infectious periods, as well as the ways they overlap, vary for different diseases.

Compartmental models

The host population is partitioned into mutually exclusive groups, referred to as compartments, based on the nature of the disease.

S: susceptible class, I: infectious class, R: recovered class

Here, we consider the simplest compartmental model called the *SIR* model. We illustrate the transmission process using the diagram, called a transfer diagram.

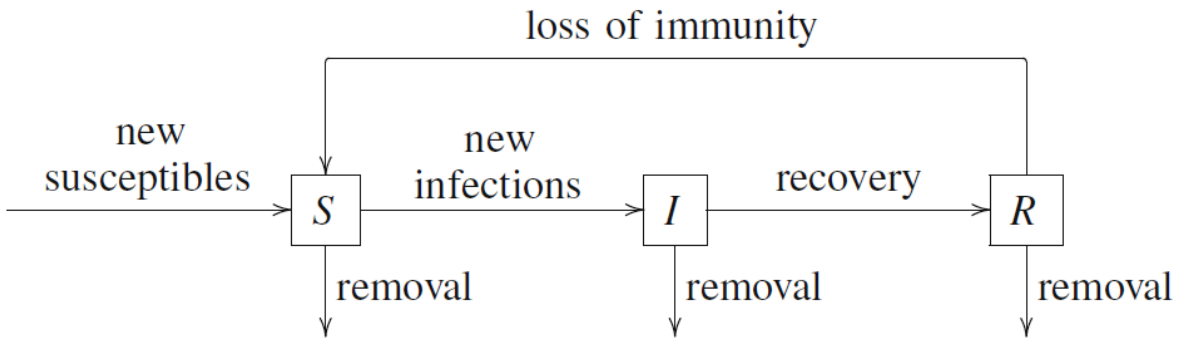

Figure: Transfer diagram for an SIR compartment model

We can construct a compartmental model by considering the net change in the number of individuals in each compartment in an interval $[t, t + \Delta t]$.

$$\begin{aligned}\Delta S(t) &= \text{new susceptibles} + \text{transfer from } R - \text{new infections} \\ &\quad - \text{removal from } S\end{aligned}$$

$$\Delta I(t) = \text{new infections} - \text{transfer into } R - \text{removal from } I$$

$$\Delta R(t) = \text{transfer from } I - \text{transfer into } S - \text{removal from } R$$

If we divide both sides of the above equations by Δt and let $\Delta t \to 0$, we obtain the following differential equations.

$$S'(t) = \text{influx of new susceptibles} + \text{transfer rate from } R$$
$$- \text{incidence rate} - \text{removal rate from } S$$

$$I'(t) = \text{incidence rate} - \text{transfer rate into } R - \text{removal rate from } I$$

$$R'(t) = \text{transfer rate from } I - \text{transfer rate into } S$$
$$- \text{removal rate from } R$$

Depending on our hypotheses about disease transmission and population transfer, the expression of the terms can vary.

Kermack-McKendrick model

Consider the following hypothesis about the transmission process of an infectious disease and its host population:

1. Transmission occurs horizontally through direct contact between hosts.
2. Mixing of individual hosts is homogeneous. In particular, the incidence rate can be expressed as $\lambda I(t)S(t)$, where λ is called the *transmission coefficient*.
3. Rate of transfer from a compartment is proportional to the population size of the compartment. For instance, the recovery rate can be written as $\gamma I(t)$, for a rate constant γ.
4. Infected individuals become infectious upon infection with no latency period.
5. There is no loss of immunity and no possibility of reinfection.
6. There is no input of new susceptibles and no removal from any compartments.
7. The total host population remains constant.

Substituting all the terms, we obtain the following system of differential equations.

$$
\begin{aligned}
S'(t) &= -\lambda IS, \\
I'(t) &= \lambda IS - \gamma I, \\
R'(t) &= \gamma I,
\end{aligned}
$$

with initial conditions

$$S(0) = S_0, \quad I(0) = I_0, \quad R(0) = 0.$$

This system of ordinary differential equations is called the Kermack-McKendrick epidemic model.

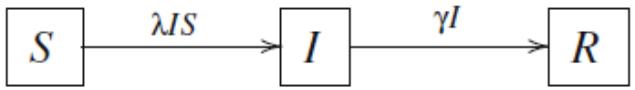

Figure: Transfer diagram for a simple SIR model

Transfer rates

Disease incidence

We are going to derive disease incidence. Let

- λ be the average per capita contact number among individuals per unit time,
- p represent the probability that a contact will result in an infection.

Then the incidence is given by

$$p\lambda \cdot \frac{S(t)}{N(t)} \cdot I(t).$$

If we combine the probability p with the contact number λ, so that λ is the per capita effective contact number, then the incidence is given by

$$\frac{\lambda}{N(t)} I(t)S(t).$$

Incidence changes depending on the nature of the total population size $N(t)$. If N is constant, then the incidence is found to be the same as the first incidence form we examined ($\beta = \frac{\lambda}{N}$). Let us consider two simple cases where the total population size $N(t)$ varies with time t.

1. The effective contact number λ is independent of the total population size. Then the incidence is given by

$$\lambda \cdot \frac{I(t)S(t)}{N(t)},$$

   where λ is a constant.

2. The effective contact number λ is proportional to the total population size, i.e., $\lambda(N) = \beta N$, for a constant β. Then the incidence is given by

$$\beta I(t)S(t).$$

Demography: birth, death, and population growth

Let's assume that the birth or death rate is proportional to the population size. Then, based on these assumptions, the corresponding system of differential equations for the model is derived.

$$
\begin{aligned}
S'(t) &= bN(t) - \lambda I(t)S(t) - d_1 S(t), \\
I'(t) &= \lambda I(t)S(t) - (\gamma + d_2)I(t), \\
R'(t) &= \gamma I(t) - d_3 R(t),
\end{aligned}
$$

where b is the natural birth rate constant, and d_1, d_2, and d_3 are death rate constants.

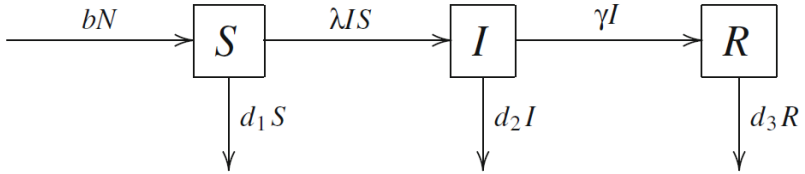

Figure: Transfer diagram for an SIR model with birth and death

Unlike previous models, for this model, the total population size changes over time.

$$N'(t) = bN(t) - d_1 S(t) - d_2 I(t) - d_3 R(t).$$

Let us consider the special case when the death rate constants for each compartment are the same (i.e., $d_1 = d_2 = d_3 = d$).

$$N'(t) = (b - d)N(t),$$

which has the general solution

$$N(t) = N_0 e^{(b-d)t},$$

where $N(0) = N_0$.

Disease latency: latent and incubation periods

Acquired immunity

Routes of transmission: horizontal and vertical

Heterogeneity

Disease control and prevention measures:

The basic reproduction number

References

- Michael Y. Li. *An Introduction to Mathematical Modeling of Infectious Diseases*. 1st ed. Springer Cham, 2018. https://doi.org/10.1007/978-3-319-72122-4.
- Porta, M. (Ed.). (2014). *A Dictionary of Epidemiology*. Oxford University Press. https://doi.org/10.1093/acref/9780199976720.001.0001